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CLASS NOTES

🤖 Module 5 · AI Automation & Agents

DAY 16 of 25

RAG: Chat With Your Own Data

"Company ke documents se jawab dene wala AI."

⚡ AAJ HUM YEH BANAYENGE

RAG Bot: Chat With Your PDFs

Do hisse: (1) Google Drive ki PDF ko chunks mein kaat ke vector store mein daalo. (2) Chat widget par sawaal pucho — agent sirf aapke documents se jawab deta hai, source ke saath. In-memory store use kiya hai taaki koi extra account na lage.

Pinecone | Qdrant | Supabase Vector | Embeddings

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

ChatGPT aapki company ke documents nahi jaanta. RAG se AI "aapke data" se jawab deta hai — sabse demanded AI project.

🎯 Aaj ke baad aap yeh kar paoge

01 RAG explained: embeddings, chunks, retrieval
🌍 Open-book exam.
⚙️ Query → embed → top-k chunks → prompt mein daalo → answer.

02 Vector stores: Pinecone, Qdrant, Supabase pgvector
🌍 Library ki almari.
⚙️ Pinecone / Qdrant / Supabase vector store nodes.

03 Document loaders & text splitters
🌍 Kitaab ko chapters mein kaatna.
⚙️ Default Data Loader + Recursive Character Splitter (chunk 800, overlap 100).

04 Ingestion pipeline: Drive → chunks → vectors
🌍 Nayi kitaab aayi → almari mein lagao.
⚙️ Drive Trigger (new file) → split → embed → insert.

05 Q&A chain with sources/citations
🌍 Jawab ke saath "page 23 dekho".
⚙️ Metadata mein file name/page → answer mein citation.

06 Evaluating & improving answer quality
🌍 Teacher copy check karti hai.
⚙️ 20 test sawaal ki sheet → answers compare → chunk size/prompt tune.

⏱ Class ka plan (2 ghante)

10m	40m	40m	20m	10m

`0:00-0:10` **Recap**
🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

`0:10-0:50` **Concept: RAG**
Library analogy + diagram
🪄 Open-book exam! Ratne ki jagah, sahi page khol ke jawab likho.

`0:50-1:30` **Live Build**
Company FAQ bot on PDFs with citations
🪄 Aaj ki kahani yaad hai — 'Library aur librarian'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

`1:30-1:50` **Challenge**
Auto-update vectors when Drive file changes
🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.

`1:50-2:00` **Q&A + Homework**
🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?

Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Embeddings

Text ka matlab numbers (vector) mein.

🪄 **Analogy:** Har shabd ka GPS coordinate

CONCEPT 02

Chunking

Lambe document ke chhote tukde (500–1000 chars, overlap).

🪄 **Analogy:** Kitaab ke paragraphs

CONCEPT 03

Vector store

Similar meaning wale chunks dhoondhne wala DB.

🪄 **Analogy:** Library ka smart librarian

CONCEPT 04

Retrieval + Generation

Relevant chunks dhoondho → LLM ko do → jawab with sources.

🪄 **Analogy:** Open-book exam

🔑 Aaj ka main idea

RAG = pehle apne documents mein se relevant hissa dhoondho (Retrieval), phir LLM ko woh hissa dekar jawab banwao (Generation). Isse AI apni company ka data bolta hai, guess nahi karta.

- Company policy PDF par HR bot
- Product manual par support bot
- Course notes par student doubt bot

📖 Ek kahani se samjho: Library aur librarian

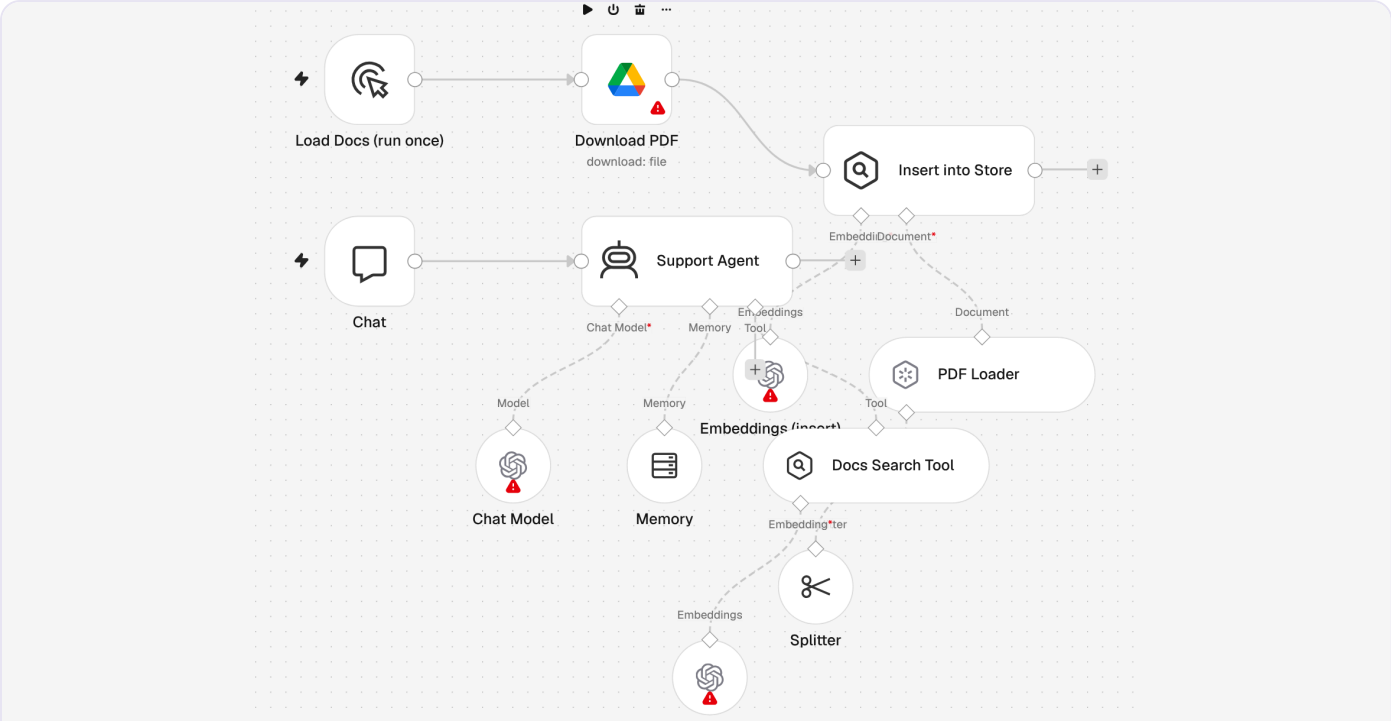
1. 📚 **Saari kitabein library mein** = Documents (Drive PDFs)
2. ✂️ **Har kitaab ke chhote panne** = Text splitter (chunks)
3. 🏷️ **Har panne par matlab ka label** = Embeddings
4. 🔍 **Librarian sahi panna dhoondhta hai** = Vector store retrieval
5. 🗣️ **Panna padh ke jawab** = LLM answer + source

🪄 Ek line mein: “Open-book exam! Ratne ki jagah, sahi page khol ke jawab likho.”

Aaj ka workflow: RAG Bot: Chat With Your PDFs

Data left se right chalta hai. Har dabba ek n8n node hai.





📸 Asli n8n mein import karne par yeh dikhega. Laal ⚠️ = account (credential) connect karna baaki.

💰 Isse kitna kama sakte ho?

Any business with FAQs/policies (RAG support bot) — aam taur par **₹25,000 – ₹75,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1 ⚡ **Google Drive Trigger**
Folder mein nayi/updated PDF

2 📁 **Google Drive**
File download

3 🧬 **Recursive Character Text Splitter**
Chunk 800, overlap 100

4 🧬 **Supabase Vector Store**
Insert documents + Embeddings OpenAI

5 ⚡ **Chat Trigger**
User question (website widget)

6 🔧 **Question and Answer Chain**
Retriever = Supabase Vector Store, answer with sources

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- [ ] Connect: **Google Drive (Google OAuth)**
- [ ] Connect: **OpenAI API key**
- [ ] Drive par ek PDF (FAQ / policy) upload karo, uski File ID paste karo
- [ ] In-memory store restart par khaali ho jaata hai — production mein Pinecone/Qdrant/Supabase vector store lagao (same wiring)

✅ Test kaise karein

1. Pehle "Load Docs" wala Test chalao
2. Phir Chat button se PDF ke baare mein sawaal pucho
3. Aisa sawaal bhi pucho jo PDF mein nahi hai — bot ko mana karna chahiye

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😩 Pehle (manual)

Customer sawaal → PDF mein dhoondhna → reply — **12 min**

12 min

⚡ Ab (n8n)

RAG bot turant source ke saath — **0.5 min**

0.5 min

⏱ 50 sawaal/din par ~10 ghante bache.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
Answer ONLY from the context. If not found, say "Mujhe iski jaankari nahi hai, main team se confirm karke batata hoon." Cite the source file.
```

📦 Sample data — input se output tak

📥 Input — 📋 Select karke copy karo

```
{
  "chatInput": "Course ki refund policy kya hai?"
}
```

📤 Output — 📋 Select karke copy karo

```
{
  "output": "Pehli 2 live classes ke baad 7 din ke andar full refund milta hai. (Source: policy.pdf, page 2)"
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 RAG agent system prompt

📋 Select karke copy karo

```
Tum company ke support assistant ho. Har jawab se pehle "company_docs" tool se search karo. Sirf wahi bolo jo documents mein hai. Agar jawab nahi mila to bolo "Mujhe iska pakka jawab nahi pata, main aapko team se connect karta hoon." Jawab ke end mein source (file/section) likho.
```

💬 Build it with ChatGPT / Claude

📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Do hisse: (1) Google Drive ki PDF ko chunks mein kaat ke vector store mein daalo. (2) Chat widget par sawaal pucho — agent sirf aapke documents se jawab deta hai, source ke saath. In-memory store use kiya hai taaki koi extra account na lage.

Flow: Manual + Drive (PDF download) → Vector Store (insert) (Chunks + embeddings) → Chat Trigger (Sawaal) → AI Agent + Store tool (Docs se jawab)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt

📋 Select karke copy karo

```
I'm building the n8n workflow "RAG Bot: Chat With Your PDFs" (nodes: Load Docs (run once), Download PDF, Insert into Store, Embeddings (insert), PDF Loader, Splitter, Chat, Support Agent, Chat Model, Memory, Docs Search Tool, Embeddings (search)).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
12 nodes · ready to import
`adifyskills.com/workflows/day-16-rag-support-bot.json`

🌐 **Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/16`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Google Drive (Google OAuth)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "OpenAI API key" connect karo.
Webhook/Form test par kuch nahi ho raha?	"Listen for test event" / Test URL use karo aur 2 minute ke andar request bhejo. Production URL sirf Active workflow par chalta hai.
AI node par 401 / 429 error?	401 = API key galat. 429 = credit khatam ya bahut jaldi requests. OpenAI billing page check karo.

⚠️ Yeh galtiyan sabse common hain

- ! Chunk size bahut bada/chhota — jawab adhoore
- ! Purane vectors delete kiye bina naye add (duplicate answers)
- ! AI ko context ke bahar jawab dene dena (hallucination)

⭐ Pro tips — time bachane wali tricks

- ★ Metadata (file name, page) save karo — citations ke liye
- ★ File update par pehle us file ke vectors delete
- ★ 20 test questions ki list banao aur har change ke baad test

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →



Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Google Drive PDFs ko ingest karke website chat widget par RAG support bot.

📝 Homework

Kisi coaching/clinic ke 5 documents par RAG bot banao aur 20 questions test karo.

☐ Homework ho gaya

BEGINNER
In-memory vector store se 1 PDF ☐

INTERMEDIATE
Drive update par auto re-index ☐

ADVANCED
Answer quality eval with test set ☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Bot har jawab ke end mein "Kya aur kuch poochna hai? 😊" likhe aur jawab 80 words se lamba na ho.

💡 Hint 1
Yeh rules agent ke system prompt mein jaate hain.

💡 Hint 2
System message ke end mein 2 rules aur jodo.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Text ko vectors mein badalne wala?

- (A) Embeddings
- (B) Parser
- (C) Memory
- (D) Merge

Q2. RAG mein "R"?

- (A) Regex
- (B) Retrieval
- (C) Random
- (D) Reply

Q3. Hallucination kam karne ke liye?

- (A) Bada font
- (B) Kuch nahi
- (C) "Only from context" instruction
- (D) Temperature 2

Q4. Lambe PDF ko todne wala?

- (A) Sort
- (B) Limit
- (C) IF
- (D) Text Splitter

📖 Aaj ke naye shabd

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

LLM
Large Language Model — ChatGPT/Claude jaisa AI jo text samajhta aur likhta hai.

RAG
AI ko pehle aapke documents mein se sahi hissa dhoondh ke dena, phir jawab banwana — open-book exam.

Vector Store
Embeddings rakhne ka database — AI ki library.

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✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 17

Document AI, Vision & Voice Automation

"Invoices padho, images samjho, voice notes transcribe karo."

☐ **Day 16 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → A
Embeddings

Q2 → B
Retrieval

Q3 → C
"Only from context" instruction

Q4 → D
Text Splitter

✅ Challenge ka solution

Support Agent → System Message mein jodo: "- Jawab max 80 words. - End mein likho: Kya aur kuch poochna hai? 😊"

📚 Official docs

- RAG in n8n ↗
- Vector stores ↗
- Embeddings ↗
- Q&A Chain ↗
- Text splitters ↗

▶ Videos

- n8n rag chatbot tutorial ↗
- n8n supabase vector store ↗

💬 Doubt poochho
WhatsApp: +91 83838 91716
wa.me/918383891716 ↗

🌐 Day 16 lesson page
Quiz, copy buttons, videos
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